

Getting Started with PIPeX: Analysis of a Multiplexed Pancreatic Cancer Dataset

Introduction

Welcome to PIPeX! This guide walks through the complete analysis of a multiplexed pancreatic cancer tissue core included with the demo dataset. By the end of this tutorial, you will have generated a high-quality cell segmentation, quantified protein expression at single-cell resolution, identified cell populations, and explored the results interactively in TissUUmaps.

This guide is intended for researchers with little or no programming experience. No prior knowledge of image analysis is required. The tutorial focuses on running a complete analysis using the recommended PIPeX workflow. More advanced options are described in the full documentation.

Before you begin

Before starting, ensure that:

- PIPeX has been installed successfully.
- The demo dataset has been downloaded.
- All image files are located in a single data directory.
- Marker names are included in the image filenames (for example, DAPI.tif or PanCK.tif).

For this tutorial, we recommend using the graphical user interface (GUI).

Workflow overview

The analysis consists of five simple steps:

1. (Optional) Preprocess the images.
2. Segment individual cells.
3. Perform downstream single-cell analysis.
4. Export the results for visualization.
5. Explore the results in TissUUmaps.

Each step can be run independently and generates quality-control outputs that allow you to verify the results before proceeding. To reproduce settings from the demo dataset, please refer to the log files included in the demo folder.

Step 1. (Optional) Image preprocessing

Why this step?

Preprocessing can improve image quality by reducing uneven illumination, background signal, and tile artifacts before segmentation. For well-balanced images, preprocessing is often unnecessary.

For your first analysis, we recommend using the default preprocessing settings unless your images contain obvious background or stitching artifacts.

After preprocessing, PIPΣX automatically saves the processed images together with quality-control images that allow you to compare the original and processed data.

Step 2. Segment individual cells

Why this step?

Segmentation identifies individual cells in the tissue and measures protein expression within every cell. These measurements form the basis of all downstream analyses.

For the demo dataset use:

- Nuclear marker: DAPI
- Membrane marker: β -catenin
- Nuclei diameter: 18 pixels
- Membrane diameter: 30 pixels
- Membrane compactness: 0.9

All remaining segmentation parameters can be left at their default values.

The nuclei and membrane diameters are the only parameters that typically require adjustment between datasets. They can be estimated by measuring representative cells using any image analysis software such as GIMP, Fiji/ImageJ, etc. An approximate estimate is usually sufficient because PIPΣX automatically adapts the segmentation around the specified diameter.

During segmentation, PIPΣX measures every selected marker for every cell and stores the results in `cell_data.csv`.

For each marker, PIPΣX automatically calculates several complementary expression scores, including mean intensity, 90th percentile intensity, the fraction of pixels above threshold, and a Gaussian Mixture Model (GMM)-based probability. The default settings are appropriate for most analyses and generally do not require adjustment.

After segmentation, review the generated quality-control images before proceeding.

Step 3. Perform downstream analysis

Why this step?

This step transforms single-cell measurements into biologically meaningful results.

PIPEX can perform normalization, dimensionality reduction, clustering, cluster refinement, and spatial neighborhood analysis using the measured protein expression profiles. For the demo dataset, the default analysis parameters are recommended.

The analysis generates:

- normalized single-cell data,
- clustering results,
- marker statistics,
- publication-ready figures,
- AnnData objects for downstream analysis.

Step 4. Export the results

Generate GeoJSON and TissUMaps outputs to visualize the segmentation and clustering results interactively.

These files can also be imported into QuPath or used for downstream analyses.

Step 5. Explore the results

Open the generated TissUMaps project.

You can now:

- visualize all measured markers,
- inspect individual cells,
- explore clustering results,
- verify segmentation quality,
- interactively browse the tissue.

Congratulations! You have completed your first PIPEX analysis.